

Long Nguyen Khac

Robotics Researcher · Control · Autonomous Systems

📍 Hanoi, Vietnam | ✉ nk.long723@gmail.com | 🏠 longnk0702.github.io | 📱 longnk0702 | 🌐 longnk0702 | 🎓 Google Scholar

Education

Vietnam National University, Hanoi – International School

Hanoi, Vietnam

B.Eng. in Automation and Informatics

Sep. 2021 – Dec. 2026

- **GPA:** 3.71/4.00; ranked in the top 4% of the cohort.
- Completed a one-year gap period for international research and exchange programs in Indonesia, Taiwan and Singapore.
- **Research focus:** robotics, wireless power transfer, and AI.

Research Experience

Water & Energy Research Lab, School of EEE, Nanyang Technological University

Singapore

Research Fellow

Feb. 2026 – Present

- **Project:** Advanced 3D Wireless Power Transfer System for Autonomous Underwater Vehicles.
- Investigating relay-coil-enhanced spatial wireless power transfer for AUV charging, with emphasis on power transfer efficiency, coil configuration, and system-level integration.
- Optimizing the mechanical design of AUVs to support underwater operation, wireless charging integration, and long-endurance autonomy.
- Exploring underwater communication between the transmitter and receiver sides of the system for reliable AUV charging coordination.

AISC Lab, AIM-HI Institute, National Chung Cheng University

Chiayi, Taiwan

Research Intern

May 2025 – Jul. 2025

- **Project:** Smart Sensing Technology for Smart Manufacturing.
- Developed a data-driven approach to estimate dynamic cutting forces in milling using 12 large-scale datasets.
- Combined nonlinear system analysis with sparse identification methods to model and predict cutting force behavior.
- Achieved moderate prediction accuracy with R^2 between 0.5 and 0.6 compared with ground-truth measurements.

AISC Lab, AIM-HI Institute, National Chung Cheng University

Remote

Remote Research Student

Aug. 2025 – Present

- **Project:** Prediction of cutting force model using AI and data-driven methods.
- Implementing machine learning models, including Random Forest and XGBoost, using 27 large-scale datasets collected through a Taguchi-designed experiment.
- Developing data-driven modeling pipelines for feature engineering, model training, and cutting force prediction in machining processes.

Intelligent Control and Artificial Intelligence Laboratory, VNUIS

Hanoi, Vietnam

Student Researcher

Sep. 2022 – Present

- **Project:** Adaptive Fuzzy Logic in Backstepping Control for 3-DOF Parallel Robot.
 - Designed the mechanical model and physical prototype of a 3-DOF delta robot, and implemented adaptive fuzzy backstepping control for robust trajectory tracking under uncertainties.
 - Reduced joint trajectory deviations to less than 0.15 rad, with stability reached in 0.15–0.2 s in simulation studies.
- **Project:** Detection of Personal Protective Equipment in Construction Sites.
 - Built a dataset of 2,632 construction-site images across five classes and trained YOLOv5s with multiple activation functions for PPE detection.

Publications

1. Long Nguyen Khac et al. *Adaptive Fuzzy Logic in Backstepping Control for 3-DOF Parallel Robot*. International Conference on Advances in Information and Communication Technology, Springer, 2024. DOI: 10.1007/978-3-031-80943-9_57.
 - **Main highlights:** an adaptive fuzzy logic control system integrated with backstepping control for a 3-DOF delta parallel robot.
 - **Presentation:** Mechatronics and Automation section in The 3rd International Conference ICTA 2024.
2. Long Nguyen Khac et al. *YOLOv5s Model with Applied Activation Functions for Personal Protective Equipment Detection in Construction Sites*. Proceedings of the National Conference on Smart Technology Applications in Industry 4.0, Smart City, and Sustainability, 2023.
 - **Main highlights:** a deep learning-based approach using YOLOv5s with six activation functions and a new dataset of 2,632 construction-site images for PPE detection.
 - **Presentation:** National Conference on Smart Technology Applications in Industry 4.0, Smart City, and Sustainability, 2023.

Teaching Experience

Faculty of Engineering and Technology, VNUIS

Hanoi, Vietnam

Teaching Assistant

Sep. 2023 – Dec. 2024

- **Course:** Modeling and Simulation of Control Systems.
- Assisted 30 students across 7 laboratory sessions on MATLAB/Simulink for robotics, mechatronics, and process control applications.
- Supported the preparation of laboratory manuals, tutorial exercises, and simulation-based learning materials.
- **Course:** Microprocessors and Microcontrollers.
- Mentored 30 students through 15 laboratory sessions on Embedded C, Arduino, STM32, Keil uVision, Arduino IDE, and STM32CubeIDE.

Hung STEM Robotics Lab

Hanoi, Vietnam

Robotics Teaching Collaborator

Jul. 2022 – Dec. 2022

- Mentored students for the World Robot Olympiad 2022 Vietnam and supported the development of LEGO-based robotics modules.
- Delivered hands-on STEM robotics sessions using LEGO NXT for students at Grammar Newton International Bilingual School, Hanoi.

Working Experience

ETEK Automation Solutions JSC

Hanoi, Vietnam

Engineering Intern

Mar. 2025 – May 2025

- Project: AGV-based Virtual Simulation for Automated Warehouse Logistics – VINFAST Hai Phong.
- Sketched AGV/AMR traffic intersections using AutoCAD and simulated warehouse logistics operations using Visual Components Premium 4.6.
- Monitored AGV/AMR behavior, identified simulation errors, and supported troubleshooting of automated logistics workflows.

Vietnam E-Commerce and Digital Economy Agency, Ministry of Industry and Trade

Hanoi, Vietnam

Technical Collaborator

Jul. 2023 – Sep. 2023

- Project: Consultancy Support and Sales Policy Development for E-Commerce Website.
- Coordinated with SMEs across Dak Nong and An Giang provinces to support e-commerce platform development.
- Assisted in website design, content management, and preparation of technical manual documents.

International Experience

NTU Singapore Global Connect Fellowship

Singapore

Research Fellowship

Feb. 2026 – Mar. 2026

- Selected as one of 39 young scholars worldwide for a fully funded research fellowship at Nanyang Technological University.
- Conducted research on advanced 3D wireless power transfer systems for autonomous underwater vehicles.
- Received the Most Popular Poster Award at the final research poster presentation.

DiscoverNUS, National University of Singapore

Singapore

Academic Exchange

Aug. 2025 – Dec. 2025

- Selected as one of 8 excellent VNU students for a fully funded exchange program at NUS.
- Expanded academic exposure to international engineering education, research culture, and interdisciplinary learning.

Taiwan Experience Education Program, National Chung Cheng University

Taiwan

Research Internship

May 2025 – Jul. 2025

- Selected as one of 28 global students for a fully funded research internship supported by the Ministry of Education, Taiwan.
- Worked on smart sensing technology and data-driven modeling for smart manufacturing.

ASEAN in Today's World, Kyushu University

Indonesia

International Study Program

Feb. 2025

- Selected as one of 4 VNU representatives to participate in the ASEAN-focused international study program.

Temasek Foundation SCALE

Vietnam & Singapore

Leadership Exchange Program

Feb. 2024

- Selected as one of 20 VNU representatives for a fully funded leadership and community action exchange program.

Honors & Awards

2026	Fully Funded Scholarship , NUS Enterprise Summer Programme in Entrepreneurship, NUS ASEAN SPARK Fellowship	<i>Singapore</i>
2026	Exchange Scholarship , Kyushu University Spring-Semester Exchange Scholarship, Department of Mechanical Engineering	<i>Japan</i>
2026	Most Popular Poster Award , NTU Singapore Global Connect Fellowship Final Research Poster Presentation	<i>Singapore</i>
2026	Research Fellowship , Selected as one of 39 young scholars worldwide for NTU Singapore Global Connect Fellowship	<i>Singapore</i>
2025	Fully Funded Scholarship , DiscoverNUS Scholarship, National University of Singapore	<i>Singapore</i>
2025	Research Internship Scholarship , Taiwan Experience Education Program, Ministry of Education	<i>Taiwan</i>
2025	International Program , ASEAN in Today's World, Kyushu University	<i>Indonesia</i>
2024	Scholarship , TOSHIBA Scholarship, TOSHIBA International Foundation	<i>Vietnam</i>
2024	Fully Funded Exchange Program , Temasek Foundation Specialist's Community Action & Leadership Exchange	<i>Vietnam & Singapore</i>
2022	Fighting Spirit Prize , South-East Asia Circuit and System Society Hackathon	<i>Vietnam</i>
2021–2025	Academic Scholarship , Six-semester recipient for excellent academic performance	<i>VNU</i>

Skills

Research	Robotics, autonomous systems, intelligent control, wireless power transfer, smart sensing, AI-enabled engineering systems
Control	Mathematical modeling, trajectory tracking, PID control, adaptive control, backstepping control, fuzzy logic control, MATLAB/Simulink simulation
Robotics	AUV systems, delta robot, AGV/AMR simulation, mechanical design, embedded system integration
Programming	MATLAB, Python, Embedded C/C++, PLC programming
Software	MATLAB/Simulink, SolidWorks, Visual Components, AutoCAD, PyTorch, TIA Portal, STM32CubeIDE, Keil uVision
AI/Data	Machine learning, data-driven modeling, Random Forest, XGBoost, YOLOv5, dataset preparation, model evaluation
Languages	Vietnamese: Native; English: IELTS 6.5, Writing 7.0; Chinese: Beginner

References

Asst. Prof. Yun Yang

Nanyang Technological University, Singapore

Assistant Professor, School of Electrical and Electronic Engineering

- Email: yun.yang@ntu.edu.sg
- Research area: power electronics, wireless power transfer, electric vehicles, renewable energy technologies.

Prof. Her-Terng Yau

*National Chung Cheng
University, Taiwan*

Director, Advanced Institute of Manufacturing with High-tech Innovations

- Email: yauherterng@gmail.com
- Research area: automatic control, mechatronics, smart manufacturing, signal processing and big data analysis, AI.

Dr. Le Xuan Hai

Vietnam National University, Hanoi

Vice Dean, Faculty of Engineering and Technology, VNUIST

- Email: hailx@vnu.edu.vn
- Research area: robotics, adaptive control, AI.